SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED

LOCATION: Jetty-1	**AREA: B**
PERMIT TO WORK No.: -	**TYPE OF SCAFFOLD:** Cantilever
SCAFFOLD DRAWING No: PMS-WA-ARCO-039	
RISK CLASSIFICATION: ✓ **High** ○ Medium ○ Low	

SCAFFOLD TO BE INSPECTED BY PIC BEFORE TAGGING

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of a Cantilever scaffold required for maintenance of the Fender at Jetty-1. The scaffold envelope is 6.8m x 2.2m x 7.3m (length x width x height) and is configured with standards, ledgers, transoms, bracing, and platforms as indicated on the approved drawing.

SAFE WORKING LOAD (SWL): 1.5 kN/m²

Total Horizontal Load in the X Direction (F_X): 2.103 kN
Total Horizontal Load in the Z Direction (F_Z): 4.807 kN

Role	ID	Name	Signature	Date
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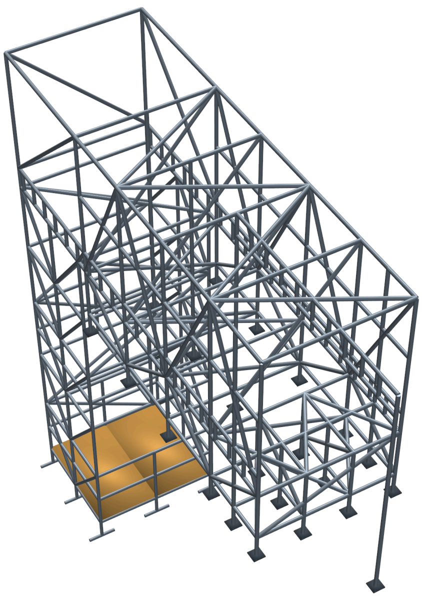
CALCULATION SHEETS

3D ISOMETRIC VIEW



GENERAL NOTE

BS EN 12811-1, 4.2.1.2 / 10.3.2.2

Scaffold tube material has a minimum nominal yield strength of 235 N/mm² (BS EN 12811-1, 4.2.1.2). A partial safety factor γ_M = 1.1 (BS EN 12811-1, 10.3.2.2) has been applied, giving a design resistance of 213.6 N/mm² used in member capacity checks. This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 - Design of Steel Structures).

Material Properties

48.3 mm outside diameter x 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

Modelling Philosophy

BS EN 12811-1, 10.2.3.2 / 10.2.3.3

A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint (KFX/KFZ = 21.18 kN). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot.

LOAD CASES

Dead Load / Self-weight

Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Live Load on Platforms

BS EN 12811-1

Platform live loading is applied to the scaffold working platforms as member UDLs in the STAAD model. The load intensity is multiplied by the tributary width of each supporting member to obtain the applied member line load. (LC 2).

Member	Tributary Width (m)	Load Intensity (kN/m²)	Member Load (kN/m)
Members 324, 327, 331, 332 (Edge)	0.800 / 2 = 0.400	1.500	**0.600**
Members 334, 397 (Interior)	(0.400 + 0.400) = 0.800	1.500	**1.200**
*Load Intensity = **1.50 kN/m²** — BS EN 12811-1 Class 2*			

Member Load = Load Intensity × Tributary Width. Edge members carry half the adjacent bay width; interior members carry half the bay from each side.
Governing load intensity (SWL) = **1.5 kN/m²** — BS EN 12811-1 Class 2.
Total applied platform live load = **4.32 kN**.

Handrail Load

Handrail loads were applied where required by the scaffold configuration and modelled independently in the global X and Z directions.

Handrail Load (HLX) = **1.08 kN** (member UDL 0.15 kN/m)
Handrail Load (HLZ) = **0.96 kN** (member UDL 0.15 kN/m)

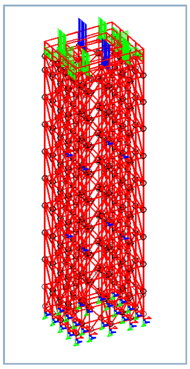
Wind Load

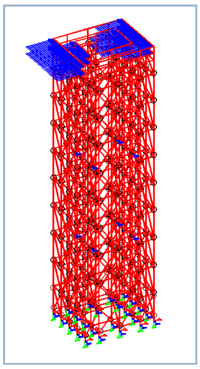
EN 1991-1-4:2005

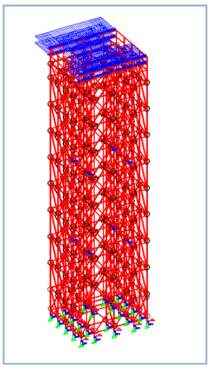
Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.064134 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

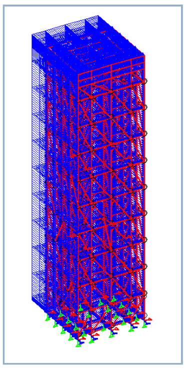
Total Wind Load - X Axis (WLX) = **11.42 kN**
Total Wind Load - Z Axis (WLZ) = **12.35 kN**

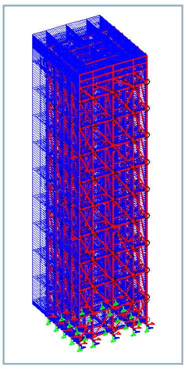
Load Diagrams


Live Load (LL)


Handrail Load X (HLX)


Handrail Load Z (HLZ)


Wind Load X (WLX)


Wind Load Z (WLZ)

WIND LOAD CALCULATION - EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	V3 =	**36.0**	m/s
	Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity		
Durst, C.S. (1960)	Convert 3-second gust to 10-minute mean wind velocity		
	V3 / V3600 =	1.52	
	V600 / V3600 =	1.06	
	V600 = (V600/V3600) x (V3600/V3) x V3 =	**25.11**	m/s
	Basic wind speed V_b = V600 =	**25.11**	m/s
EN 1991-1-4:2005	Roughness Length (z0) - Terrain Category 0	0.003	m
	Terrain category II (z0,ii)	0.05	
EN 1991-1-4, Table 4.1	z_{min}	1.0	m
	z - height of structure under consideration	**7.3**	m
EN 1991-1-4, Sec 4.3	$C_{r0}(z)$ - Roughness factor	1.0	

Reference	Calculation	Value	Unit
EN 1991-1-4, Sec 4.3	$C_o(z)$ - Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor $K_r = 0.19 \times [z0/z0,ii]^{0.07}$	0.156	
EN 1991-1-4, Table 4.1	$z_{min} < z$ -> Case 1 applies		
	Wind Turbulence		
	$C_r(z) = K_r \times \ln(z/z0)$	1.2166	
	Mean velocity $V_m(z) = C_r(z) \times C_o(z) \times V_b$	30.543	m/s
EN 1991-1-4, Sec 4.4	K1	1.0	
	Turbulence intensity $I_v(z) = k1 / [C_o(z) \times \ln(z/z0)]$	0.1283	
	Peak Velocity Pressure		
	Density of air rho	1.25	kg/m³
EN 1991-1-4, Sec 4.5	$q_p(z) = [1 + 7{*}I_v(z)] \times 0.5 \times rho \times V_m(z)^2$	**1106.5235**	N/m²
Wind Load = C_f x q_p x A_f			
C_f =		1.2	
q_p =		1.106523	kN/m²
A_f =	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		**0.064134**	**kN/m**

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use gamma = **1.5**; SLS (characteristic) combinations use gamma = **1.0**.

DL = Dead Load | **LL** = Platform Live Load | **HLX/HLZ** = Handrail X/Z | **WLX/WLZ** = Wind X/Z

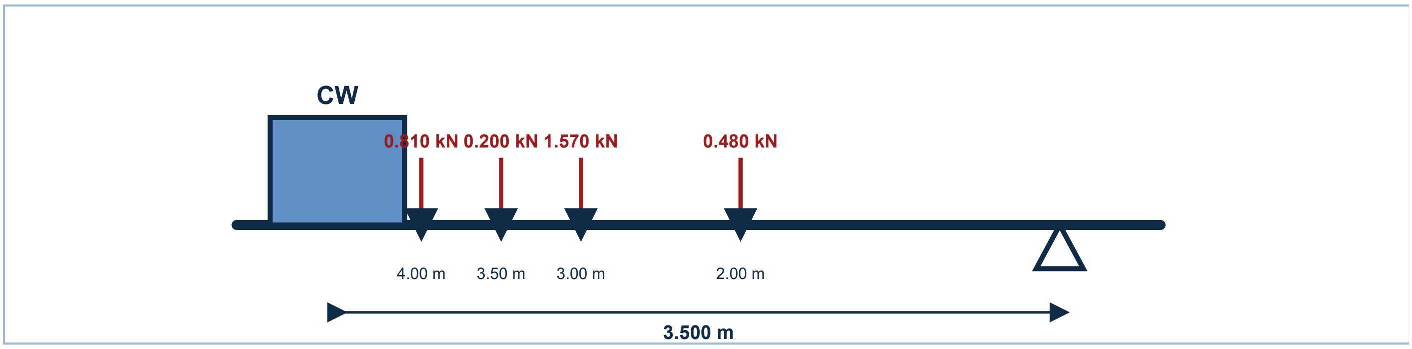
LC No.	Combination	Component Loadings & Factors
7	**SLS DL + LL**	LC1 x 1.0 + LC2 x 1.0
8	**SLS DL + LL + HLX**	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0
9	**SLS DL + LL + HLZ**	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0
10	**SLS DL + WLX**	LC1 x 1.0 + LC5 x 1.0
11	**SLS DL + WLZ**	LC1 x 1.0 + LC6 x 1.0
12	**SLS DL + LL + WLX**	LC1 x 1.0 + LC2 x 1.0 + LC5 x 1.0
13	**SLS DL + LL + WLZ**	LC1 x 1.0 + LC2 x 1.0 + LC6 x 1.0
14	**SLS DL + LL + WLX + HLX**	LC1 x 1.0 + LC2 x 1.0 + LC3 x 1.0 + LC5 x 1.0
15	**SLS DL + LL + WLZ + HLZ**	LC1 x 1.0 + LC2 x 1.0 + LC4 x 1.0 + LC6 x 1.0
16	**ULS 1.5DL + 1.5LL**	LC1 x 1.5 + LC2 x 1.5
17	**ULS 1.5DL + 1.5LL + 1.5HLX**	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5
18	**ULS 1.5DL + 1.5LL + 1.5HLZ**	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5
19	**ULS 1.5DL + 1.5WLX**	LC1 x 1.5 + LC5 x 1.5
20	**ULS 1.5DL + 1.5WLZ**	LC1 x 1.5 + LC6 x 1.5
21	**ULS DL + 1.5WLX**	LC1 x 1.0 + LC5 x 1.5
22	**ULS DL + 1.5WLZ**	LC1 x 1.0 + LC6 x 1.5
23	**ULS 1.5DL + 1.5LL + 1.5WLX**	LC1 x 1.5 + LC2 x 1.5 + LC5 x 1.5
24	**ULS 1.5DL + 1.5LL + 1.5WLZ**	LC1 x 1.5 + LC2 x 1.5 + LC6 x 1.5
25	**ULS 1.5DL + 1.5LL + 1.5WLX + 1.5HLX**	LC1 x 1.5 + LC2 x 1.5 + LC3 x 1.5 + LC5 x 1.5
26	**ULS 1.5DL + 1.5LL + 1.5WLZ + 1.5HLZ**	LC1 x 1.5 + LC2 x 1.5 + LC4 x 1.5 + LC6 x 1.5

COUNTERWEIGHT STABILITY CHECK

Counterweight Load Calculation

From STAAD.Pro analysis, the following values were obtained as force acting on the cantilever base.

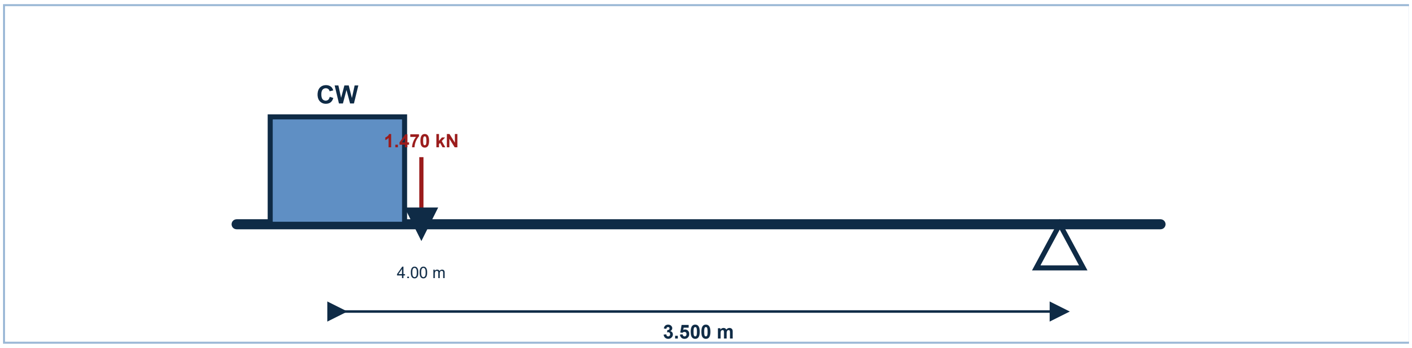
Counter weight Analysis A




```
0.81 x 4.0 + 0.48 x 2.0 + 0.2 x 3.5 + 1.57 x 3.0
= 9.61 kN
CW = 9.61 / 3.5
```

CW = **2.746 kN**

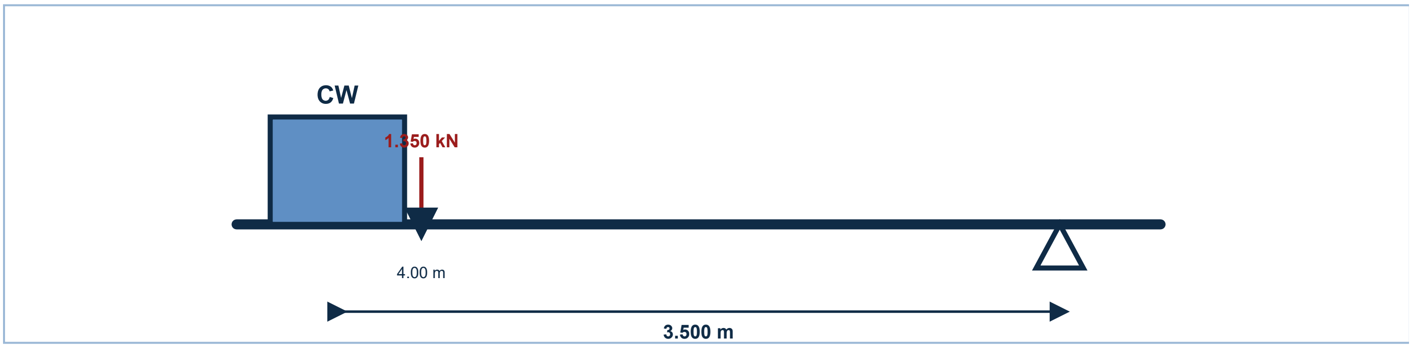
Counter weight Analysis B




```
1.47 x 4.0
= 5.88 kN
CW = 5.88 / 3.5
```

CW = **1.68 kN**

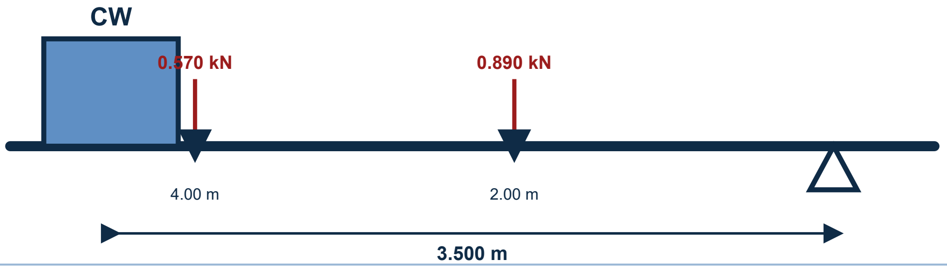
Counter weight Analysis C




```
1.35 x 4.0
= 5.4 kN
CW = 5.4 / 3.5
```

CW = **1.543 kN**

Counter weight Analysis D




```
0.89 x 2.0 + 0.57 x 4.0
= 4.06 kN
CW = 4.06 / 3.5
CW = 1.16 kN
```

Summation

```
Sum counter weight A, B, C, D = 2.746 kN + 1.68 kN + 1.543 kN + 1.16 kN = 7.129 kN


Use factor of safety of 4.0
Total counterweight = 7.129 kN x 4.0 = 28.514 kN


Use counterweight of 3.0 tonnes (29.43 kN).
```

STATIC CHECK PARAMETERS - UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.544 at Member No. **30**, Load Combination **26**, Clause **EC-6.3.3-662**. **PASS** - All members adequate; UC ratio below unity.

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
30	PIP483.6	PASS	EC-6.3.3-662	**0.544**	26	16.11	C
31	PIP483.6	PASS	EC-6.2.9.1	**0.494**	26	2.86	C
45	PIP483.6	PASS	EC-6.2.9.1	**0.482**	26	0.56	C
293	PIP483.6	PASS	EC-6.2.9.1	**0.454**	25	0.98	C
50	PIP483.6	PASS	EC-6.3.3-662	**0.377**	26	16.71	C
47	PIP483.6	PASS	EC-6.2.9.1	**0.375**	26	4.99	C
46	PIP483.6	PASS	EC-6.2.9.1	**0.374**	26	3.54	C
546	PIP483.6	PASS	EC-6.2.9.1	**0.367**	26	7.83	C
534	PIP483.6	PASS	EC-6.2.9.1	**0.358**	26	0.41	C
253	PIP483.6	PASS	EC-6.2.9.1	**0.344**	25	5.70	T
439	PIP483.6	PASS	EC-6.2.9.1	**0.281**	26	0.00	C
257	PIP483.6	PASS	EC-6.2.9.1	**0.270**	25	7.69	T
32	PIP483.6	PASS	EC-6.2.9.1	**0.260**	26	4.17	C
29	PIP483.6	PASS	EC-6.2.9.1	**0.259**	26	1.68	T
3	PIP483.6	PASS	EC-6.2.9.1	**0.256**	26	3.03	T
25	PIP483.6	PASS	EC-6.2.9.1	**0.250**	26	5.67	T
71	PIP483.6	PASS	EC-6.3.3-661	**0.248**	26	2.72	C
97	PIP483.6	PASS	EC-6.3.3-662	**0.246**	26	18.28	C
44	PIP483.6	PASS	EC-6.2.9.1	**0.235**	26	0.65	C
353	PIP483.6	PASS	EC-6.3.3-662	**0.231**	26	0.57	C

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
51	PIP483.6	PASS	EC-6.2.9.1	**0.230**	26	0.70	T
255	PIP483.6	PASS	EC-6.2.9.1	**0.229**	25	2.00	T
572	PIP483.6	PASS	EC-6.3.3-661	**0.220**	16	6.53	C
387	PIP483.6	PASS	EC-6.2.9.1	**0.216**	26	0.19	T
443	PIP483.6	PASS	EC-6.2.9.1	**0.213**	26	0.30	T
339	PIP483.6	PASS	EC-6.2.9.1	**0.212**	26	1.59	T
514	PIP483.6	PASS	EC-6.3.3-661	**0.205**	26	9.45	C
554	PIP483.6	PASS	EC-6.3.3-661	**0.204**	26	6.03	C
455	PIP483.6	PASS	EC-6.2.9.1	**0.201**	26	0.03	T
573	PIP483.6	PASS	EC-6.3.3-661	**0.201**	21	6.18	C
427	PIP483.6	PASS	EC-6.2.9.1	**0.196**	26	0.52	T
98	PIP483.6	PASS	EC-6.3.3-662	**0.195**	25	6.19	C
532	PIP483.6	PASS	EC-6.2.9.1	**0.195**	26	2.20	T
511	PIP483.6	PASS	EC-6.2.9.1	**0.193**	26	4.15	T
246	PIP483.6	PASS	EC-6.2.9.1	**0.190**	26	3.45	C
334	PIP483.6	PASS	EC-6.2.9.1	**0.185**	25	0.37	T
447	PIP483.6	PASS	EC-6.3.3-662	**0.184**	26	0.21	C
67	PIP483.6	PASS	EC-6.3.3-662	**0.181**	26	1.76	C
397	PIP483.6	PASS	EC-6.2.9.1	**0.181**	18	0.41	T
449	PIP483.6	PASS	EC-6.2.9.1	**0.181**	26	2.63	T
36	PIP483.6	PASS	EC-6.2.9.1	**0.171**	25	0.48	C
37	PIP483.6	PASS	EC-6.2.9.1	**0.171**	25	0.50	C
38	PIP483.6	PASS	EC-6.2.9.1	**0.171**	25	0.19	C
39	PIP483.6	PASS	EC-6.2.9.1	**0.171**	25	0.23	C
40	PIP483.6	PASS	EC-6.2.9.1	**0.171**	25	0.88	C
41	PIP483.6	PASS	EC-6.2.9.1	**0.171**	25	1.48	C
28	PIP483.6	PASS	EC-6.2.9.1	**0.170**	25	1.04	T
33	PIP483.6	PASS	EC-6.2.9.1	**0.170**	25	1.72	T
34	PIP483.6	PASS	EC-6.2.9.1	**0.170**	25	1.36	T
35	PIP483.6	PASS	EC-6.2.9.1	**0.170**	25	1.49	T
Showing highest 50 utilization ratios; 445 lower-utilization members omitted.							

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **9.970 kN** (Member 314, C). The required right-angle coupler class is selected from Table C.1 below using the slipping resistance limit.

Selected coupler class: **Class A** (capacity = 10.0 kN).

Check: 9.970 kN <= 10.0 kN -> **PASS**

Class A right-angle couplers are adequate.

Table C.1 - Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force $F_{s,k}$ (kN)	**10.0**	**15.0**
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.8
	Pull-apart force $F_{p,k}$ (kN)	20.0	30.0
	Rotational moment $M_{T,k}$ (kNm)	-	0.13
Swivel Coupler (SW)	Slipping force $F_{s,k}$ (kN)	6.0	9.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.8
	Pull-apart force $F_{p,k}$ (kN)	15.0	22.5
	Rotational moment $M_{T,k}$ (kNm)	-	0.08
Putlog / Single Coupler (PL)	Slipping force $F_{s,k}$ (kN)	-	3.0
	Cruciform bending moment $M_{B,k}$ (kNm)	-	0.6

Member Axial Force Summary (Top 30 Horizontal Members — ULS Code Check)

Member	Type	Status	Clause	UC Ratio	L/C	Axial Force (kN)
314	C	PASS	EC-6.3.3-661	0.131	26	**9.970**
514	C	PASS	EC-6.3.3-661	0.205	26	**9.450**
284	C	PASS	EC-6.3.3-662	0.153	26	**9.270**
278	C	PASS	EC-6.3.3-662	0.130	26	**8.200**
546	C	PASS	EC-6.2.9.1	0.367	26	**7.830**
272	C	PASS	EC-6.3.3-662	0.121	26	**7.150**
265	T	PASS	EC-6.2.9.1	0.076	25	**6.790**
512	C	PASS	EC-6.3.3-661	0.151	26	**6.750**
572	C	PASS	EC-6.3.3-661	0.220	16	**6.530**
266	C	PASS	EC-6.3.3-662	0.113	26	**6.270**
573	C	PASS	EC-6.3.3-661	0.201	21	**6.180**
554	C	PASS	EC-6.3.3-661	0.204	26	**6.030**
271	T	PASS	EC-6.2.9.1	0.059	25	**5.740**
253	T	PASS	EC-6.2.9.1	0.344	25	**5.700**
25	T	PASS	EC-6.2.9.1	0.250	26	**5.670**
260	C	PASS	EC-6.3.3-662	0.110	26	**5.390**
247	C	PASS	EC-6.3.3-662	0.112	26	**5.330**
241	C	PASS	EC-6.3.3-662	0.076	26	**5.130**
235	C	PASS	EC-6.3.3-662	0.058	26	**5.030**
548	C	PASS	EC-6.3.3-662	0.082	26	**4.540**
507	C	PASS	EC-6.3.3-661	0.103	26	**4.310**
270	T	PASS	EC-6.2.9.1	0.069	25	**4.290**
544	C	PASS	EC-6.3.3-661	0.112	26	**4.260**
511	T	PASS	EC-6.2.9.1	0.193	26	**4.150**
471	C	PASS	EC-6.3.3-662	0.052	26	**3.900**
472	C	PASS	EC-6.3.3-662	0.052	26	**3.860**
240	C	PASS	EC-6.3.3-662	0.127	26	**3.450**
246	C	PASS	EC-6.2.9.1	0.190	26	**3.450**
261	C	PASS	EC-6.2.9.1	0.167	25	**3.370**
234	C	PASS	EC-6.3.3-662	0.077	26	**3.310**

FRICTIONAL RESISTANCE

Max vertical reaction per support (Max f_y) = Resistance x 0.15 / C_F	= **10.59 kN**
Coefficient of friction (C_F)	= **0.3**
Frictional Resistance (FR) = C_F x Max f_y	= **3.177 kN**
Spring Stiffness = FR / 0.15	= **21.18 kN** <- applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state — values from STAAD.Pro SLS (unfactored) combination results. Displacement tables are shown below.

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement - LC 15 Member 415 | Member length L = 200.0 mm	**0.798 mm**	L/100 = 200/100 = **2.0 mm**	PASS

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
1557	0.790	0.750	15	1897	0.120	0.074	9.322
1560	0.787	0.750	15	1906	0.119	0.074	9.403
675	0.657	0.417	15	1522	0.132	-0.688	4.444
1571	0.630	0.750	15	2382	-0.214	-0.851	7.999
1568	0.608	0.750	15	2468	-0.214	-0.859	7.983
675	0.578	0.417	9	1728	-0.040	-0.856	1.110
675	0.548	0.500	13	1824	0.070	-0.710	3.284
675	0.546	0.500	14	1829	0.389	-0.707	3.332
1509	0.527	0.875	15	2845	0.121	0.078	6.486
1560	0.518	0.750	9	2893	-0.096	-0.248	4.614
1512	0.518	0.875	15	2896	0.120	0.079	6.533
1557	0.514	0.750	9	2918	-0.095	-0.257	4.552
678	0.511	0.583	15	1957	0.175	-0.540	2.721
672	0.504	0.583	15	1982	0.083	-0.529	2.621

Max = **0.798 mm** (LC 15) | L = Member 415 length (200 mm) | Allowable L/100 = 2.0 mm → PASS

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement - LC 15 Member 405 | Member length L = 300.0 mm	**0.425 mm**	L/200 = 300/200 = **1.5 mm**	PASS

Horizontal Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
647	0.425	0.667	15	2352	0.125	0.110	3.898
647	0.342	0.667	9	2921	-0.090	-0.217	0.714
650	0.337	0.667	15	2970	0.121	0.090	2.288
646	0.332	0.667	15	3013	0.121	0.090	2.195
587	0.303	1.167	15	6600	0.136	0.118	4.108
590	0.275	1.167	15	7279	0.139	0.107	2.401
586	0.273	1.167	15	7314	0.130	0.104	2.320
1500	0.268	0.208	15	1868	-0.239	-0.828	4.462
647	0.260	0.583	14	3848	0.313	-0.040	2.980
647	0.259	0.583	13	3855	0.029	-0.052	2.939
587	0.246	1.167	13	8142	0.049	-0.036	3.162
587	0.243	1.167	14	8215	0.310	-0.027	3.182
590	0.236	1.167	14	8489	0.314	-0.059	1.895
650	0.235	0.667	14	4264	0.318	-0.078	1.789

Max = **0.425 mm** (LC 15) | L = Member 405 length (300 mm) | Allowable L/200 = 1.5 mm → **PASS**

CONCLUSION

The structural analysis and design of the **Cantilever** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.544** at Member 30). Vertical and horizontal deflections are within their respective allowable limits of **L/100 = 2.0mm** and **L/200 = 1.5mm**. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

INSTALLATION NOTES

1. The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.
2. Confirm that the Cantilever scaffold configuration matches the approved drawing and is suitable for maintenance.
3. The work vicinity must be barricaded and appropriate signage placed at all access points.
4. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
5. A valid work permit must be obtained, discussed, and signed before erection commences.
6. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
7. Install and secure cantilever support members, needles, anchors, ties, and back-span restraints exactly as shown on the approved drawing before building the working platform.
8. Build the scaffold frame to the approved envelope of 6.8m x 2.2m x 7.3m, checking level, plumb, and cantilever projection at each lift.
9. Install bracing, ledgers, transoms, platform boards, guardrails, and toe boards progressively, maintaining all specified ties and restraints.
10. Install mid-level horizontal ledgers and transoms at 0.8m height in accordance with the working drawing.

11. Install mid-level horizontal ledgers and transoms at 1.3m height in accordance with the working drawing.
12. Install mid-level horizontal ledgers and transoms at 2.3m height in accordance with the working drawing.
13. Install mid-level horizontal ledgers and transoms at 2.8m height in accordance with the working drawing.
14. Install mid-level horizontal ledgers and transoms at 3.3m height in accordance with the working drawing.
15. Install mid-level horizontal ledgers and transoms at 4.3m height in accordance with the working drawing.
16. Install mid-level horizontal ledgers and transoms at 4.6m height in accordance with the working drawing.
17. Install mid-level horizontal ledgers and transoms at 5.3m height in accordance with the working drawing.
18. Install mid-level horizontal ledgers and transoms at 5.6m height in accordance with the working drawing.
19. Install mid-level horizontal ledgers and transoms at 6.5m height in accordance with the working drawing.
20. Install mid-level horizontal ledgers and transoms at 7.0m height in accordance with the working drawing.
21. Install all plan, face, and longitudinal bracing shown on the approved drawing to provide sway resistance in both principal directions.
22. Install platform boards, guardrails, mid-rails, toe boards, access arrangements, and any handrail load-resisting members required by the design.
23. Tighten and torque-check all load-bearing couplers and support connections before the scaffold is released for inspection.
24. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before use.
25. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
26. All modifications made after erection must be carried out by a competent scaffold team with the approval of the scaffold engineer. End-user or third-party modification of this scaffold structure is strictly prohibited.
27. For dismantling, perform the erection steps in reverse order.

REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment - Scaffolds - Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures - General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures - Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design — Load Combinations, Partial Factors, and Limit States (ULS/SLS/EQU)	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Durst, C.S.	Wind Speeds Over Short Periods of Time. The Meteorological Magazine, 89, 181–186 — basis of 3-second gust to 10-minute mean conversion	1960
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 - Structural Analysis and Design Software	2022

Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.
Report: Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.